1. Use the quadratic formula to solve the equation.

$x^2 + 3x - 1 = 0$

x = [ ]
(Simplify your answer. Type an exact answer, using radicals as needed. Use a comma to separate answers as needed.)

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2. The U.S. Crude Oil production, in billions of barrels, for the years from 2010 projected to 2030, can be modeled $y = -0.001x^2 + 0.058x + 1.968$, with x equal to the years after 2010 and y equal to the number of billions of barrels of crude oil.

a. Find and interpret the vertex of the graph of this model.
b. What does the model predict the crude oil production will be in 2033?
c. Graph the function for the years 2010 to 2030.

a. The vertex of the graph of this model is v = ([ ], [ ]). (Round to three decimal places as needed.)

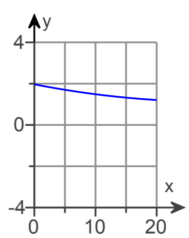
Choose the correct interpretation of the vertex below.

- **A.** The minimum number of barrels of crude oil projected to be produced during this period is 29.000 billion barrels.
- **B.** The maximum number of barrels of crude oil projected to be produced during this period is 29.000 billion barrels.
- **C.** The minimum number of barrels of crude oil projected to be produced during this period is 2.809 billion barrels.
- **D.** The maximum number of barrels of crude oil projected to be produced during this period is 2.809 billion barrels.

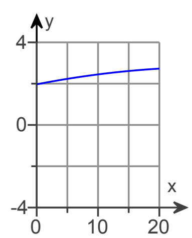
b. The model predicts the crude oil production to be [ ] billion barrels of oil in 2033. (Round to three decimal places as needed.)

c. Choose the correct graph below.

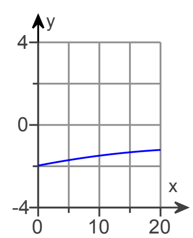
A.



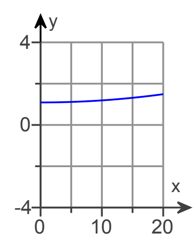
B.



C.



D.



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3. The total revenue function for a product is given by R = 360x dollars, and the total cost function for this same product is given by $C = 6000 + 40x + x^2$, where C is measured in dollars. For both functions, the input x is the number of units produced and sold.
a. Form the profit function for this product from the two given functions.
b. What is the profit when 15 units are produced and sold?
c. What is the profit when 29 units are produced and sold?
d. How many units must be sold to break even on this product?

a. Write the profit function.
P(x) = [ ]
(Simplify your answer.)

b. When 15 units are produced and sold, there is a

(1) ________ of $[ ].
(Simplify your answer.)

c. When 29 units are produced and sold, there is a

(2) ________ of $[ ].
(Simplify your answer.)

d. The number of units that must be sold to break even on this product is [ ].
(Simplify your answer. Use a comma to separate answers as needed.)

(1) ○ profit ○ loss

(2) ○ loss ○ profit

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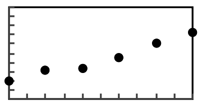
4. Partially because of people from a certain country living longer, the number with Alzheimer's disease and other dementia is projected to grow each year. The table below gives the millions of citizens in the country with Alzheimer's from 2000 and projected to 2050. Complete parts (a) through (e).

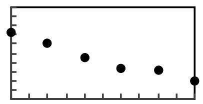
Year	2000	2010	2020	2030	2040	2050
Number	3.9	6.1	6.6	8.9	12.1	14.6

a. Create a scatter plot of the data, with x equal to the number of years after 2000 and y equal to the number of millions of citizens with Alzheimer's disease or other dementia. Choose the correct graph below.

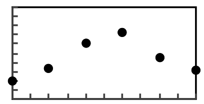
A.



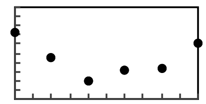
B.



C.



D.



All graphs are shown in [0,50] by [0,20] with Xscl = 5 and Yscl = 2 viewing rectangles.

b. Find a quadratic function that models y, the number of millions of people with Alzheimer's disease or other dementia, as a function of x, the number of years after 2000. Report the model with four significant digits.

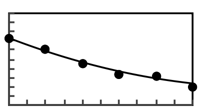
$y = (\_\_\_)x^2 + (\_\_\_)x + \_\_\_$

(Type integers or decimals.)

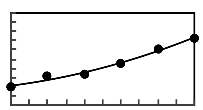
c. Graph the data and the model on the same axes, and comment on the fit.

First graph the data and the model on the same axes. Choose the correct graph below.

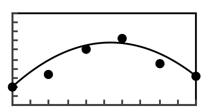
A.



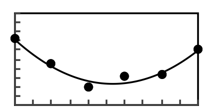
B.



C.



D.



All graphs are shown in [0,50] by [0,20] with Xscl = 5 and Yscl = 2 viewing rectangles.

Comment on the fit of the model on the data. Choose the correct answer below.

- **A.** The model is an excellent fit because it passes through all of the data points.
- **B.** The model is a poor fit because it does not pass close to any of the data points.
- **C.** The model is a mediocre fit because it passes close to some of the data points.
- **D.** The model is a good fit because it passes close to all of the data points.

d. How many citizens does the model predict will have Alzheimer's disease or other dementia in 2058?

___ million

(Type an integer or decimal rounded to one decimal place.)

e. During what year after 2000 does the model predict that 10.3 million citizens will have Alzheimer's or other dementia?

___

5. Suppose the weekly cost for the production and sale of bicycles is $C(x) = 26x + 4306$ dollars and that the total revenue is given by $R(x) = 82x$ dollars, where x is the number of bicycles.

   a. Write the equation of the function that models the weekly profit from the production and sale of x bicycles.
   b. What is the profit on the production and sale of 200 bicycles?
   c. Write the function that gives the average profit per bicycle.
   d. What is the average profit per bicycle if 200 are produced and sold?

   a. $P(x) =$ ☐

   b. The profit is \$☐.

   c. $\overline{P}(x) =$ ☐

   d. The average profit per bicycle is \$☐. (Round to the nearest cent as needed.)

6. A manufacturer of a particular item has monthly fixed costs of \$600 and variable costs of \$10 per item, and it sells the items for \$20 per item.
   a. Write a function that models the monthly profit P from the production and sale of x units of the item.
   b. What is the profit if 110 items are produced and sold in 1 month?
   c. At what rate does the profit grow as the number of items increases?

   The function that models the monthly profit P from the production and sale of x units of the item is P = ☐.
   (Simplify your answer. Do not factor.)

   The profit of 110 items produced and sold in 1 month is \$☐.
   (Simplify your answer.)

   The rate at which the profit grows is \$☐ per item.
   (Simplify your answer.)

*7. Suppose that a person's heart rate x minutes after vigorous exercise has stopped can be modeled by $f(x) = \frac{4}{5}(x-10)^2 + 90$. The output is in beats per minute, where the domain of f is $0 \le x \le 10$. **(a)** Evaluate f(0) and f(3). Interpret the result. **(b)** Estimate the times when the person's heart rate was between 100 and 120 beats per minute, inclusive.

   (a) $f(0) =$ ☐ and $f(3) =$ ☐
   (Simplify your answer. Type an integer or decimal.)

   Choose the correct interpretation.

   ○ **A.** Rate is 170 bpm initially and about 130 bpm after 3 min.
   ○ **B.** Rate is 130 bpm initially and about 170 bpm after 3 min.

   (b) The times when the person's heart rate was between 100 and 120 beats per minute is about ☐ $\le x \le$ ☐ (min).
   (Simplify your answer. Type an integer or decimal rounded to one decimal place as needed.)

*8. Find f[g(x)] and g[f(x)].

   $$f(x) = \frac{x}{6} + 7, \quad g(x) = 2x - 1$$

   f[g(x)] = ☐

   g[f(x)] = ☐

*9. Find f(x) and g(x) such that $h(x) = (f \circ g)(x)$ and $g(x) = 8 - 7x$.

$h(x) = (8 - 7x)^3 - 2(8 - 7x)^2 + 7(8 - 7x) - 1$

f(x) = [ ]

10. Consider the table giving values for the forward and inverse functions.

$f(5) = -6$

$f^{-1}(3) = -7$

$f(9) = -7$

$f(3) = 2$

$f^{-1}(5) = 2$

$f^{-1}(9) = -6$

Find the values of the following.

$f(-6) =$ [ ]

$f^{-1}(-6) =$ [ ]

$f(-7) =$ [ ]

$f^{-1}(-7) =$ [ ]

$f(2) =$ [ ]

$f^{-1}(2) =$ [ ]

11. For many species of fish, the weight W is a function of the length x, given by $W = kx^2$, where k is a constant depending on the species. Suppose k = 0.04, W is in pounds, and x is in inches, so the weight is $W(x) = 0.04x^2$.

a. Find the inverse function of this function.
b. What does the inverse function give?
c. Use the inverse function to find the length of a fish that weighs 9 pounds.

a. Give the inverse function.

$W^{-1}(x) =$ [ ]

b.

The input units for W^{-1} are

- **A.** pounds
- **B.** inches

The output units for W^{-1} are

- **A.** pounds
- **B.** inches

c. A fish that weighs 9 pounds is [ ] inches long.